

Visual Studio Code interface showing a Python script for image processing using OpenCV and NumPy. The script reads a sample image, converts it to grayscale, and applies Sobel edge detection. The results are displayed in four windows: Grayscale Image, Sobel-X Output, Sobel-Y Output, and Combined. The script also saves the combined edge detection results to a file named 'edges.png'.

```
1 import cv2
2 import numpy as np
3 image = cv2.imread("sample.jpg")
4 grey_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
5
6 sobel_x = np.array(
7     [
8         [-1, 0, 1],
9         [-2, 0, 2],
10        [-1, 0, 1]
11     ]
12 )
13
14 sobel_y = np.array(
15     [
16         [-1, -2, -1],
17         [0, 0, 0],
18         [1, 2, 1]
19     ]
20 )
21
22 output1 = cv2.Sobel(grey_image, cv2.CV_64F, 1, 0, ksize=3)
23 output2 = cv2.Sobel(grey_image, cv2.CV_64F, 0, 1, ksize=3)
24 sobel_combined = cv2.magnitude(output1.astype(np.float32), output2.astype(np.float32))
25 sobel_combined = cv2.normalize(sobel_combined, None, 0, 255, cv2.NORM_MINMAX)
26 print(sobel_combined.dtype, sobel_combined.max())
27
28 cv2.namedWindow("Grayscale Image", cv2.WINDOW_NORMAL)
29 cv2.namedWindow("Sobel-X Output", cv2.WINDOW_NORMAL)
30 cv2.namedWindow("Sobel-Y Output", cv2.WINDOW_NORMAL)
31 cv2.namedWindow("Combined", cv2.WINDOW_NORMAL)
32
33 cv2.imwrite("edges.png", cv2.convertScaleAbs(sobel_combined))
34 cv2.imshow("Grayscale Image", grey_image)
35 cv2.imshow("combined", cv2.convertScaleAbs(sobel_combined))
36 cv2.imshow("Sobel-X Output", cv2.convertScaleAbs(output1))
37 cv2.imshow("Sobel-Y Output", cv2.convertScaleAbs(output2))
38 cv2.waitKey(0)
39 cv2.destroyAllWindows()
```

Terminal output:

```
float32 255.0
PS D:\coding\Assignment 4>
```